VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING (CO3117)

Assignment Report

Text Classification on AG News

Instructor: Dr. Le Thanh Sach

Members:

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Contents

1 Course Context

- Course: Machine Learning (CO3117)
- Department: Computer Science and Engineering, HCMUT – VNU-HCM
- Supervisor: Dr. Le Thanh Sach
- Topic: Text Data Classification (Topic 2)

2 Project Objectives

The project aims to build a complete text classification pipeline from exploratory data analysis to final model comparison, following the required workflow:
1. EDA and data understanding.
2. Configurable preprocessing and TF-IDF feature extraction.
3. Classical ML training and evaluation (Logistic Regression, SVM, Naive Bayes).
4. Modern embedding extension with SBERT.
5. Comparison across feature families with report-ready artifacts.

3 Dataset and Problem Setting

- Dataset: `ag_news` from Hugging Face Datasets.
- Task: 4-class news topic classification.
- Split used in full mode: 120,000 training samples and 7,600 test samples.

4 Source Code and Reproducibility

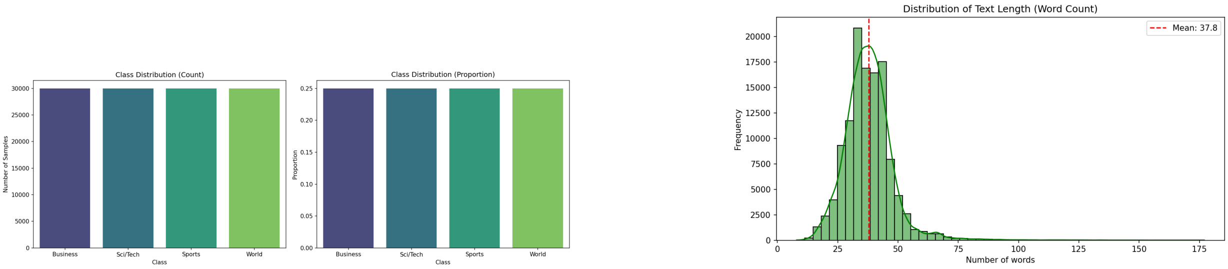
The complete source code, artifacts, and instructions to run this project are available on our public GitHub repository:

https://github.com/THVKhang/MachineLearning_TextModule

All executions are designed to be reproducible via a single Colab Notebook entry point, without requiring personal cloud storage mounting.

5 Exploratory Data Analysis (EDA)

Extensive Exploratory Data Analysis was performed to understand the characteristics of the AG News dataset.



(a) Class Distribution

(b) Text Length Distribution

Figure 1: Overview of Dataset Distributions

Key findings include:

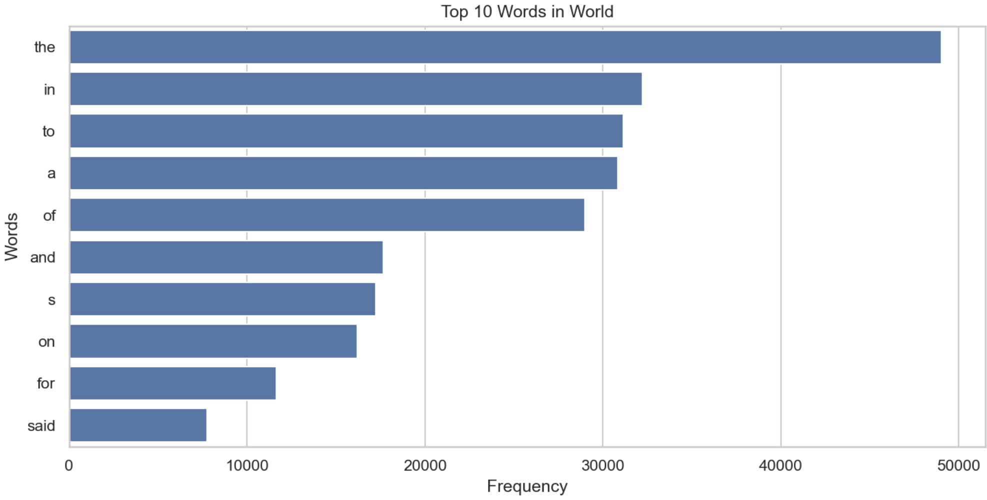
– **Balanced dataset:** The 120,000 training samples are perfectly balanced across the 4 classes (30,000 each). This eliminates the need for class weighting or resampling techniques.

– **Short texts:** The average text length is approximately 35 words. Most semantic content can be captured effectively using TF-IDF with bigrams.

– **Domain-specific vocabulary:** Each class has distinct top keywords (e.g., 'game" for Sports, 'stock" for Business).

– **Low noise:** The dataset is very clean, with near-zero empty or duplicate texts.

– **Minimal URLs:** Only about 32 URL-heavy texts were found in the training set, having negligible impact on overall quality.


Top 10 Words in World
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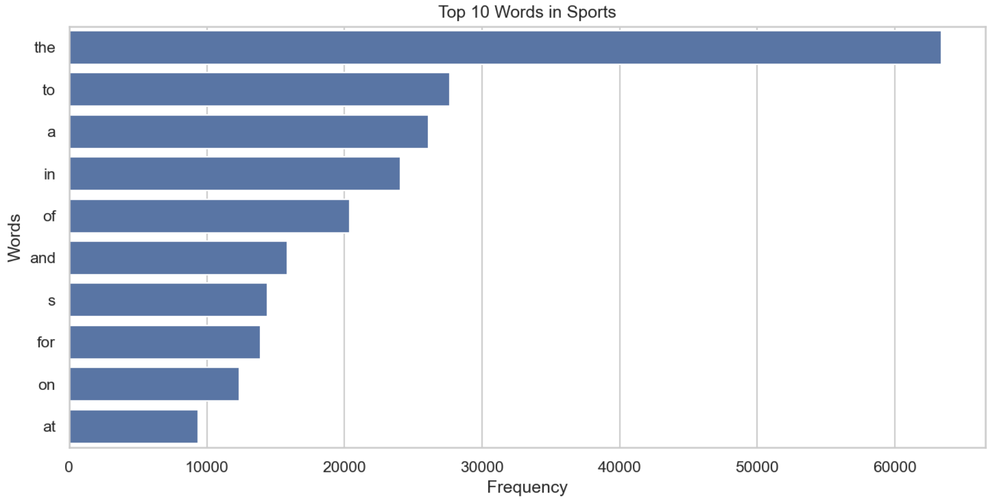
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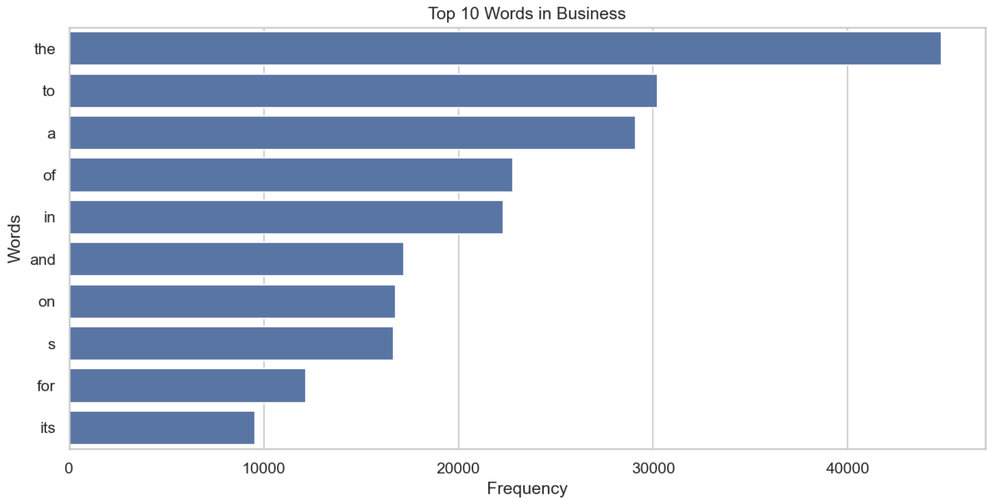
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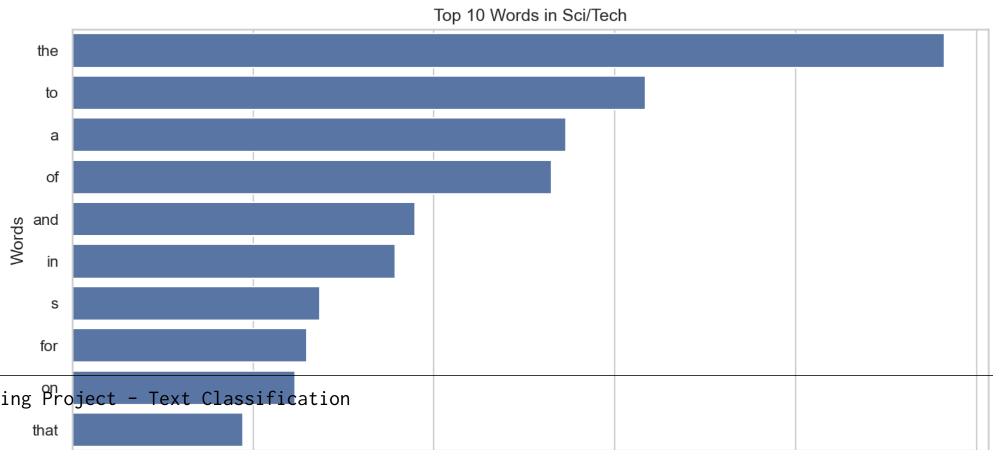
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6 Implemented Methodology

6.1 Traditional Pipeline

– **Preprocessing:** Data cleaning, handling missing values, and tokenization (lowercasing, stopword removal, punctuation removal, number removal).

– **Feature Extraction:** Traditional methods evaluated include TF-IDF configuration search over three candidates: `tfidf_uni_1k`, `tfidf_uni_bi_3k`, and `tfidf_uni_bi_5k`.

– **Evaluation:** Performance metrics evaluated include Accuracy, Precision, Recall, and F1-score (weighted). Primary model-selection metric: F1-weighted.

– **Model Training:** Supervised Learning algorithms utilized include Logistic Regression, SVM, and Naive Bayes.

6.2 Modern Embedding Extension

– **Feature Extraction:** Modern Contextual Embeddings evaluated using `sentence-transformers/all-MiniLM` (a BERT-based model).

– Benchmark scales: `5k_2k` and `20k_2k`.

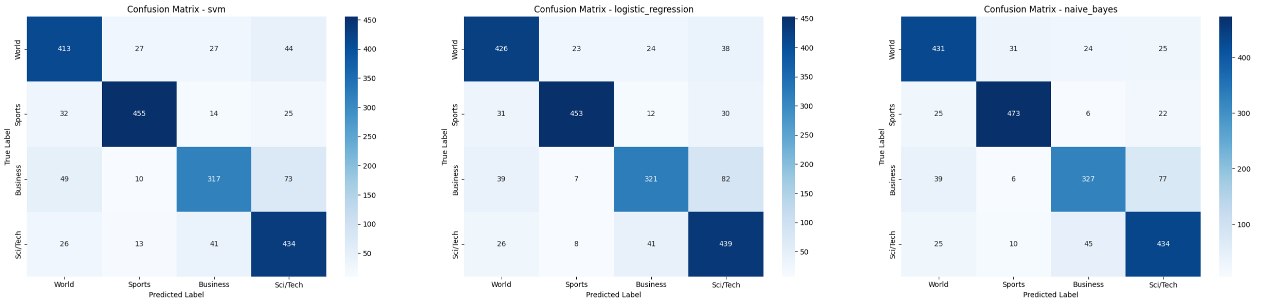
– Numerical Features: All embeddings and labels are exported and serialized as `.npy` artifacts.

7 Results

7.1 Best TF-IDF Results (Full Data)

Model	Accuracy	Precision (w)	Recall (w)	F1 (w)
SVM	0.9046	0.9044	0.9046	0.9044
Logistic Regression	0.9038	0.9036	0.9038	0.9036
Naive Bayes	0.8871	0.8866	0.8871	0.8867

Table 1: TF-IDF model comparison.



(a) SVM (b) LogReg (c) Naive Bayes

Figure 3: Confusion Matrices for Traditional Models

7.2 SBERT Benchmark Results

Scale	Best Model	Accuracy	Precision (w)	Recall (w)	F1 (w)
5k_2k	Logistic Regression	0.8740	0.8734	0.8740	0.8736
20k_2k	SVM	0.8955	0.8958	0.8955	0.8954

Table 2: Best embedding model per scale.

7.3 Feature-Family Comparison

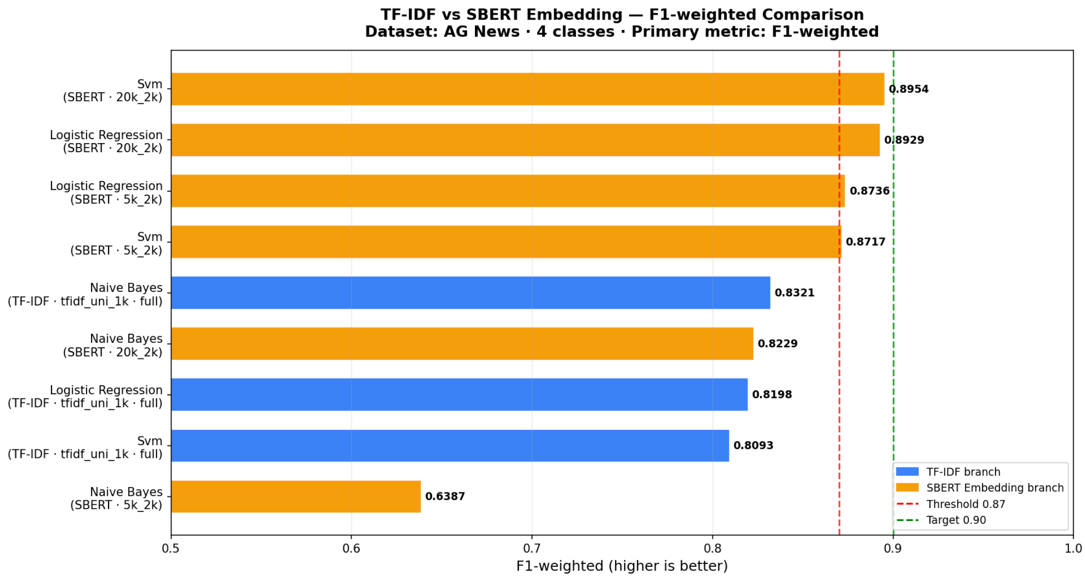


Figure 4: Performance Comparison: TF-IDF vs SBERT Embedding

The final Comparative Analysis shows that the best TF-IDF model (SVM with F1 = 0.9044) marginally outperformed the best Transformer-based contextual embedding model (SVM at 20k scale with F1 = 0.8954). The Transformer-based model underperformed the traditional SVM baseline in this specific context because the AG News dataset consists of very short texts (average 35 words) with highly distinct, domain-specific vocabularies (e.g., "stock", "baseball"). Traditional TF-IDF excels at capturing these explicit keyword signals without requiring complex semantic relationships. Furthermore, TF-IDF is significantly faster to train and extract, while maintaining highly interpretable sparse features. Therefore, TF-IDF + SVM is the primary recommendation.

8 Error Analysis

An error analysis was performed on the best TF-IDF model (SVM) to understand the distribution of misclassifications. Out of 7,600 test samples, the SVM model made 381 errors.

The top 3 most common confusion pairs are:

1. **True: Business → Predicted: Sci/Tech** (73 occurrences)
2. **True: Business → Predicted: World** (49 occurrences)
3. **True: World → Predicted: Sci/Tech** (44 occurrences)

This indicates a semantic overlap between Business/Technology news and Business/World news, which is common in articles discussing international tech corporations or global markets.

9 Team Contributions

Member	ID	Email	Main Contribution	%
Tran Hoang Vy Khang	2352502	2352502@hcmut.edu.vn	Embeddings, workflow, integration	30
Le Tan Minh Khoa	2352563	2352563@hcmut.edu.vn	Text cleaning and TF-IDF	20
Tao Nguyen Quang Khang	2352499	2352499@hcmut.edu.vn	ML training and evaluation	20
Tran Gia Huy	2252264	2252264@hcmut.edu.vn	Pipeline and config architecture	15
Vo Le Hai Dang	2352257	2352257@hcmut.edu.vn	Data loading and EDA analysis	15

Table 3: Detailed contribution table with student emails.